

A Small Visit to the Wightman Field Presentation

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1 Wightman Field Presentation

Fundamental data

$$(\mathcal{H}, \Omega, U, \mathcal{D}, \{\hat{\Phi}_A\}_{A \in I})$$

1. $\mathcal{H}$: Hilbert space.

2. Vacuum vector

$$\Omega \in \mathcal{H}$$

vacuum state.

Like the identity element in a group?

3.

$$U$$

Unitary representation of the Poincaré group on $\mathcal{H}$.

4.

$$\mathcal{D}$$

dense common invariant domain.

$$\mathcal{D} \subset \mathcal{H}$$

A good subspace that all field operators and symmetry transformations can act on.

- dense: large enough, can approximate every vector in $\mathcal{H}$;
- invariant:

$$U(g)\mathcal{D} \subset \mathcal{D},$$

i.e. after acting with a symmetry transformation, vectors are still inside $\mathcal{D}$.

5.

$$\{\hat{\Phi}_A\}_{A \in I}$$

a collection of quantum fields.

A labels different types of fields.

All $\hat{\Phi}_A$'s are operator-valued distributions on a test function space $\mathcal{T}$.

$$\hat{\Phi}_A(f) = \int d^d x \hat{\Phi}_A(x) f(x)$$

(smearing)

Like the delta distribution,

$$\int \delta(x) f(x) dx = f(0).$$

Classical Wightman Case

$$\mathcal{T} = S(\mathbb{R}^d)$$

- S : Schwarz space.

If

$$f \in S,$$

then f and all of its derivatives decrease very rapidly at infinity.

- Therefore,

$$f \in S(\mathbb{R}^d)$$

is a good test function. It guarantees that vacuum distributions are tempered.

Domain Assumptions

$$\Omega \in \mathcal{D}, \quad U(g)\mathcal{D} \subset \mathcal{D}, \quad \hat{\Phi}_A(f)\mathcal{D} \subset \mathcal{D}, \quad \text{etc.}$$

We want every operation on vectors in $\mathcal{D}$ to stay inside $\mathcal{D}$.

$\implies$

Definition of a common invariant domain.

So finite products of field operators are well-defined.

Continuous Multilinear Functionals

Matrix elements of finite products are required to be continuous multilinear functionals of the test functions.

$$(f_1, \dots, f_n) \mapsto \left\langle \psi, \hat{\Phi}_{A_1}(f_1) \cdots \hat{\Phi}_{A_n}(f_n) \chi \right\rangle \in \mathbb{C},$$

where

$$\psi, \chi \in \mathcal{D}.$$

Hence,

$$\left\langle \psi, \hat{\Phi}_{A_1}(x_1) \cdots \hat{\Phi}_{A_n}(x_n) \chi \right\rangle$$

is a distribution.

When

$$\psi = \chi = \Omega,$$

we obtain the vacuum correlation functions

$$\left\langle \Omega, \hat{\Phi}_{A_1}(f_1) \cdots \hat{\Phi}_{A_n}(f_n) \Omega \right\rangle.$$

Covariance

Fields transform covariantly under Poincaré transformations.

Let

$$g = (a, \Lambda) \in \mathcal{P},$$

with

$$gx = \Lambda x + a.$$

Poincaré transformations act on spacetime points.

The transformed test function is

$$f_g(y) = f(g^{-1}y),$$

i.e. use g^{-1} to pull back the function.

Covariance of the field:

$$U(g)^{-1}\hat{\Phi}_A(f)U(g) = R(\Lambda)_A{}^B\hat{\Phi}_B(f_g).$$

Symmetry transformation = change Lorentz indices together with the spacetime argument.

Scalar vs. Vector Fields

For a scalar field,

$$R(\Lambda) = 1,$$

so

$$U(g)^{-1}\hat{\Phi}_A(f)U(g) = \hat{\Phi}_A(f_g).$$

For vector fields, Lorentz transformations mix different components,

$$\hat{A}_\mu \longrightarrow \Lambda_\mu{}^\nu \hat{A}_\nu,$$

so we have a nontrivial representation

$$R(\Lambda)_A{}^B.$$

Sometimes one writes the covariance directly for local fields,

$$U(g)^{-1}\hat{\Phi}_A(x)U(g) = R(\Lambda)_A{}^B\hat{\Phi}_B(g^{-1}x).$$

Spectrum Condition

$$\text{supp } E_P \subset \overline{V}_+.$$

Here

$$E_P$$

is the joint spectral measure of the translation generators. Equivalently, energy and momentum lie inside the forward light cone,

$$p^0 \geq 0, \quad p^2 \geq 0.$$

Locality

Smearing version:

$$\left[\hat{\Phi}_A(f), \hat{\Phi}_B(h) \right]_{\pm} = 0,$$

whenever

$$\text{supp}(f) \quad \text{and} \quad \text{supp}(h)$$

are spacelike separated.

Wightman Functions

The n -point Wightman function is

$$W_{A_1 \dots A_n}(x_1, \dots, x_n) = \left\langle \Omega, \hat{\Phi}_{A_1}(x_1) \cdots \hat{\Phi}_{A_n}(x_n) \Omega \right\rangle.$$

Primary correlation data = vacuum distributions.

They describe field correlations in the vacuum state. They are translationally invariant, so they only depend on relative positions.

Spectrum Condition for Wightman Functions

After Fourier transform,

$$\text{supp } \widehat{W}_{A_1 \dots A_n} \subset \overline{V}_+^{n-1}.$$

Here

$$\widehat{W}_{A_1 \dots A_n}$$

is the Fourier transform of the Wightman distribution. Only supported in the positive energy–momentum region.

Positivity

Take a finite sequence of test tensors,

$$F = \left\{ F_{A_1 \dots A_n}^{(n)} \right\}_{0 \leq n \leq N},$$

where

$$F^{(0)} \in \mathbb{C}.$$

Define

$$\Psi_F = F^{(0)}\Omega + \sum_{n=1}^N \sum_{A_1 \dots A_n} \int F_{A_1 \dots A_n}^{(n)}(x_1, \dots, x_n) \hat{\Phi}_{A_1}(x_1) \dots \hat{\Phi}_{A_n}(x_n) \Omega d^d x_1 \dots d^d x_n.$$

Construct a finite linear combination of states generated from the vacuum.

Example:

$$\Psi = c\Omega + \hat{\Phi}(f)\Omega + \hat{\Phi}(f_1)\hat{\Phi}(f_2)\Omega.$$

- first term: vacuum;
- second term: one field acting on the vacuum;
- third term: two fields acting on the vacuum.

Require

$$\langle \Psi_F, \Psi_F \rangle \geq 0.$$

Correlation functions should really come from some Hilbert space.

Vacuum Cyclicity

Starting from the vacuum, field operators generate the whole Hilbert space.

$$\overline{\text{span} \left\{ \hat{\Phi}_{A_1}(f_1) \dots \hat{\Phi}_{A_n}(f_n) \Omega : n \geq 0, f_i \in \mathcal{T}, A_i \in I \right\}} = \mathcal{H}.$$

Equivalently,

$$\Omega + \text{field operators} + \text{linear combinations} \implies \mathcal{H}.$$